

Comparison of scene contrast temperature in mid-wave infrared and long-wave infrared

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ABSTRACT. One of the primary activities in emissive infrared imager design is the trade on whether to use midwave infrared (MWIR) or longwave infrared (LWIR) in the application. Applications include target acquisition (both target search and target identification), threat warning, aircraft detection, and pilotage. There has been a great deal of work in the characterization of MWIR versus LWIR target signatures. There has been much less work in the characterization of scene (sometimes called background) contrast. The scene contrast of the background can be just as important in the performance of the sensor in an application. A few examples are: (1) a high scene contrast with high clutter contrast can make target search much more difficult, (2) a high scene contrast with image-based navigation can enhance the performance of location estimation, and (3) high scene contrast with mobility sensors can enhance the flying performance of a rotorcraft pilotage system. We discuss the differences observed in scene contrast between MWIR and LWIR bands. This provides a scene contrast characterization for emissive infrared applications. Radiometrically calibrated imagery is acquired with MWIR and LWIR cameras in various environments, and the measured MWIR and LWIR scene contrast is compared. The radiometric comparison is performed in terms of the standard deviation of the scene equivalent blackbody temperature. Comparisons are provided under different conditions, such as rural versus urban and day versus night. This comparison enables the infrared system designer with the means to perform detailed engineering trades.

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1 Introduction

Scene contrast temperature (SCT) describes the global variation of the flux in a scene [within the field of view (FOV) of a system] in radiometric terms. The standard deviation of the blackbody equivalent temperatures of the scene, σ_T , in units of Kelvin is an indicator of SCT and can be considered a local descriptor. Vollmerhausen and Jacobs defined SCT as “the delta radiometric temperature in the scene needed to generate the average display luminance when minimum luminance is zero.”¹ This definition is consistent with the fact that the thermal image arises from small variations in temperature and emissivity within the scene that rides on a large background flux.

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SCT is defined as the delta radiometric temperature corresponding to half of the dynamic range of the display and is therefore directly related to the full dynamic range of the display. For an imager viewing objects on the ground, the dynamic range will be set according to the range of signals coming from the scene. For a normally distributed scene, 99.7% of the signal is within $3\sigma_T$ of the mean, if the dynamic range of the display is set to $6\sigma_T$, 0.3% of the scene will be saturated, and $SCT = 3\sigma_T$. For a display dynamic range set to $4\sigma_T$, 5% of the scene will be saturated but the overall sensitivity of the system will increase. This trade off in display dynamic range is generally an issue of personal preference for the user (gain and level), but the SCT increases monotonically with σ_T . This trend is true regardless of the statistical distribution of signal across the scene.

The definition of SCT is based on linear contrast stretching and the assumption of a display dynamic range as perceived by an observer. This definition is a global definition of SCT. The σ_T is a local descriptor of scene contrast and, depending on the scale, can be used to discuss the effect of clutter in the scene or the effect of the scene contrast in other applications such as pilotage and global positioning system (GPS)-denied image-based navigation. In this paper, only σ_T will be measured and reported, however it is important to consider the importance of SCT in the context of applications, despite being a function of the user's preferences.

High SCT and high σ_T can cause a reduction in performance for some sensor applications, such as target detection in the wide field of view search task and target identification in narrow field of view applications. Low SCT and σ_T cause a reduction in other sensor applications such as pilotage performance or GPS-denied, image-based navigation. Low SCT and σ_T correspond to conditions where there is not much thermal variation in the scene, such as after long periods of rainy or overcast weather. High SCT and σ_T are caused by the Sun beating the objects in the scene with high thermal capacities for long periods of time. These objects will retain this solar loading over time and re-emit photons as a function of their temperature and emissivity.

Pilotage systems are designed to give operators enhanced situational awareness, especially at nighttime and in degraded visual environments. SCT provides a description of the scene contrast under conditions when pilots fly, and the conditions can vary significantly. The targeting task performance (TTP) metric is designed to quantify the performance of a targeting system but can also be used to quantify the performance of a pilotage system as a function of SCT.

A pilot can see the surroundings much better with high σ_T and therefore high SCT. Situational awareness is enhanced because the when σ_T is much greater than the sensor noise, as this will lead to a higher SCT and noise taking a very small part of the dynamic range of the display. Figure 1 shows a typical pilotage performance plot of TTP value as a function of SCT with four examples of possible pilotage images.

There are four figures near the pilotage TTP plot where the SCT of the terrestrial part of the scene (not including the cold sky) is 0.2K in the upper left, 0.5K in the lower left, 1K in the lower right, and 3K in the upper right. Each example image corresponds to a single SCT and a resulting

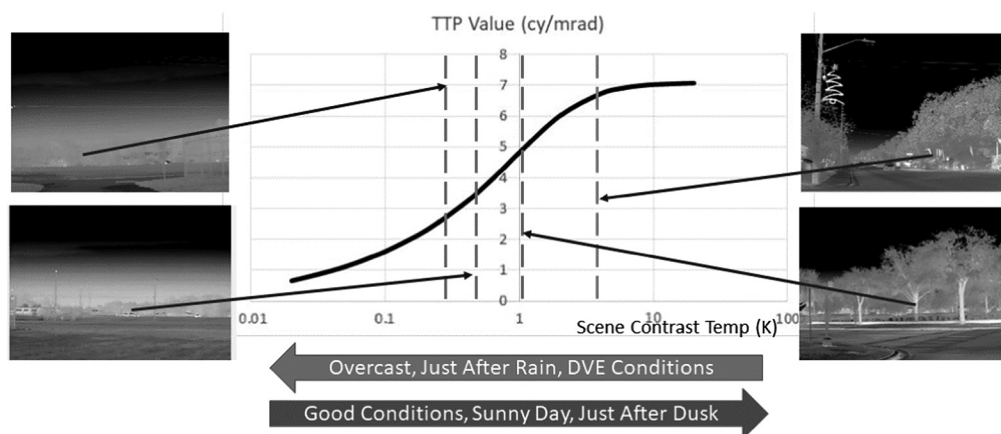


Fig. 1 Typical pilotage sensor performance curve as a function of SCT. Example images correspond to dotted lines.

TTP value as indicated by each dotted line. The black pilotage performance curve reduces with a lower SCT, and at some point, the TTP is low enough that it is not safe to fly.

The second case where a high SCT and σ_T are desirable is a recent application when ground vehicles and aircraft are in GPS-denied conditions and images of thermal imagers (e.g., uncooled microbolometers) is compared to a world or regional image database. These comparisons are used to pinpoint the location of the vehicle similar to GPS. Image correlation and deep learning techniques in autonomous systems are methods used to make these comparisons. In any of these cases, a higher scene contrast provides better signals to be compared to the image databases. Additionally, techniques, such as simultaneous localization and mapping, benefit from higher SCT and σ_T for similar reasons to pilotage and image-based navigation.

Opposite to pilotage and image-based navigation, low SCT in search and targeting is desirable. In search range performance literature,^{2,3} the target contrast seen by the sensor is

$$\text{Target contrast} = \frac{\Delta T_{\text{tgt}}}{2\text{SCT}}, \quad (1)$$

where ΔT_{tgt} is the target differential signature in Kelvin. The target contrast equation implies that a larger SCT reduces the target contrast and makes the task of target detection and target identification more difficult when the target ΔT_{tgt} remains unchanged. For target identification performance estimates (in a narrow FOV), the apparent SCT is used. The apparent SCT is the SCT propagated through the atmosphere in a manner similar to the apparent target differential signature, ΔT_{tgt} . The target and background within the FOV are adjusted to span the display dynamic range. Consider a target with a radiometric temperature difference of 1K, a σ_T of 5K (assuming a user preferring dynamic range of the display to span $6\sigma_T$) leads to a SCT of 15K and a total dynamic range of 30K, with the target spanning 1/30th of the dynamic range. For a smaller σ_T , the total dynamic range will be smaller, allowing the target to span a larger portion of the total dynamic range.

Consider the two images in Fig. 2. The image on the left has a low SCT and σ_T (less variation in scene equivalent blackbody temperature) and the image on the right has a high SCT and σ_T . For a target of low signature, such as a small differential temperature, the target would be easier to find and identify in the left scene as it would span a greater percentage of the dynamic range of the display.

While there are spatial characteristics of the scene that correspond to high clutter, the SCT describes the amplitude of the scene corresponding to the signals that compete with the target signature⁴ when linear contrast stretching is based on global contrast. Some scenes with high SCT and many objects in the scene that are close to the target size would be considered “high clutter” scenes. For these scenes, σ_T can capture the clutter complexity.

This paper provides comparisons of mid-wave infrared (MWIR) and long-wave infrared (LWIR) σ_T and indirectly SCT. Differences in σ_T occur when there are various objects in a scene that emit in-band photons at a high differential rate to the other band. The number of photons that an object emits is a function of the emissivity of the object and will change with Planck’s curve, according to temperature.

There are many factors that differentially affect the scene contrast output of LWIR and MWIR systems. The Sun emits more photons in MWIR and less photons in LWIR. MWIR

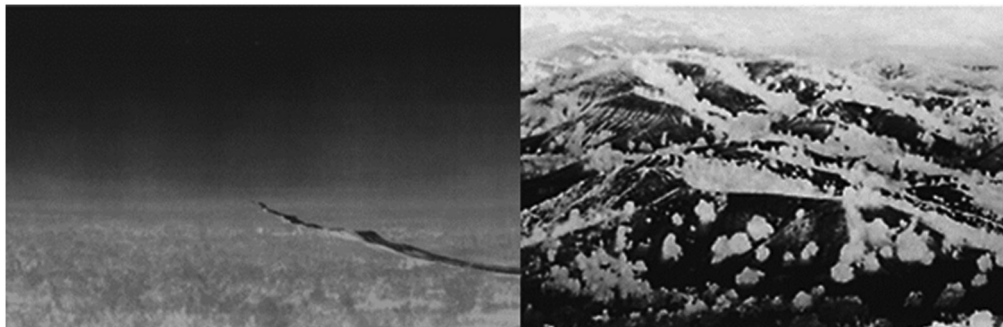


Fig. 2 The left image has low SCT, and the right image has a high SCT.

objects typically have higher reflectivity, meaning MWIR will have additional scene contrast at daytime due to solar reflections. Objects may have wavelength dependent emissivities and higher differential object emissivities may create higher scene contrast in one band. Additionally, the apparent temperature difference between the sky and terrestrial objects is higher in the LWIR than the MWIR due to a lower path radiance in the LWIR than the MWIR, meaning that horizon contrast will be greater, and reflections of the sky will create higher contrast.

In this paper, the scene contrast of MWIR and LWIR are empirically compared at daytime and nighttime in both rural and urban scenes. A diurnal study is conducted to highlight the effects of daytime on scene contrast in these bands. These comparisons are performed “up close” such that the atmosphere between the sensor and the measured scenes has negligible transmission or path radiance degradation. For a longer sensor to scene paths, an “apparent” SCT can be determined but is beyond the scope of this paper. The rest of the paper is outlined as follows: Sec. 2 provides a background on previous work and radiometry, Sec. 3 describes the experimental process including sensors, calibration, and conditions, Sec. 4 reports scene contrast data, Sec. 5 discusses significant findings in the data, and Sec. 6 draws conclusions about said findings.

2 Background

While there is not much work in the literature about quantifying the differences between MWIR and LWIR SCT, there are some small efforts that quantified clutter in the MWIR and LWIR as well as some anecdotal observations about SCT regarding pilotage. In Vollmerhausen’s design approach to pilotage sensors,⁵ in one exercise called FROSTY APACHE, the SCT contrast was observed over a period of frequent rain and snow. The SCT was very low, and the background was washed out so that it was difficult to pilot an aircraft. However, a benefit from the low background contrast was that the target acquisition performance of the targeting system was enhanced since warm targets in a washed-out background were easy to find and identify. The targets “popped out” of the background imagery.

At the Kyoto Technical University of Japan,⁶ researchers measured the SCT of city, forest, mountains, and the sky above the mountains over a period of 10 years. The measurements were taken in all seasons and under various ambient conditions. Their goal was to model infrared clutter, but the bands they selected were 2 to 3, 3 to 4, 4 to 5, and 8 to 14 micrometers, making it difficult to provide a direct MWIR to LWIR comparison. The measurements showed that the 8 to 14 micrometers band had more spatial variance than any other band. The 4 to 5 and 8 to 14 micrometer bands had consistent spatial variance in the forest regardless of cloud cover or time of day. Also, each spectral band had one or two orders of magnitude difference in variance between daytime and night for most scenes. The purpose of the study was to quantify infrared background clutter. They concluded that the emissive bands provide a Gaussian distribution, and the reflective bands provide a Poisson distribution in radiance.

Georgia Tech Research Institute provided a study⁷ that was intended to quantify background clutter⁸ in the MWIR and LWIR. They used radiometers on a helicopter looking down on the ground to obtain the data. The spectral bands were 3 to 5 and 8 to 12 micrometers. The purpose of the study was to quantify clutter in the two spectral bands and determine how weather, scene type, and measurement conditions changed the clutter. Scenes were classified as open, tree line, or mixed. The classifications were insufficient because important properties of the scene were ignored, and they used a low pass and high pass filter corresponding to the desired clutter sizes. With these caveats, there were some useful observations from the effort. The MWIR and LWIR data did not show the same behavior regarding spatial variances. The diurnal data showed that both bands peaked in the middle of the day with solar reflections and solar loading. The variations were less at night, but the variations for MWIR were more consistent than LWIR. Overall, the data showed that LWIR variance was higher than MWIR but that the MWIR was less susceptible to environmental changes. There were no measurements in precipitation related to rain or snow.

Thermal crossover occurs due to the natural diurnal cycle of ambient temperatures. Solar loading causes objects to heat up during the day and objects in the scene retain different amounts of solar radiation. After sunset, the air temperature drops and the objects cool off while emitting their solar loaded energy. Eventually, all objects are at similar temperatures and the scene has low

Table 1 Vollmerhausen and Bui results from Ref. 9.

Image at horizon	1	3	5	7	9	11	13	15	16	18
LWIR/MWIR ratio	2.5	2.6	3.5	2.4	5.3	2.8	1.8	2.3	2.0	3.5
Image below horizon	2	4	6	8	10	12	14	17	19	20
LWIR/MWIR ratio	1.0	0.8	1.1	0.9	1.1	0.9	1.1	1.7	1.1	1.2

contrast. Thermal crossover is considered a “worst case” for pilotage systems that must use this contrast for terrain-following night flights. For Vollmerhausen’s development of the TTP metric for pilotage performance involved a study,⁹ where backgrounds were segmented for two cases (boxes of interest): the horizon and below the horizon. In the study, the standard deviation of the pixel values in the box was compared for the MWIR and LWIR spectral bands. Thermal crossover was prevalent mainly for scenes that included the horizon. Table 1 below shows the results in terms of the LWIR to MWIR ratio of the standard deviation measurements.

Images below the horizon tended to provide nearly equal contrast regardless of where the images were acquired. These results did not agree with the results from Ref. 6, but the bands were quite different. A larger factor is that Vollmerhausen and Bui compared the contrasts in equivalent blackbody temperatures and the Kyoto study compared contrast in radiance. A description of the comparison between equivalent blackbody temperature, radiance, and conversion to photons/second will be provided later in this section.

While the US Army used tree-to-ground contrast as the metric for providing useful imagery,⁹ the horizon is also another important aspect of pilotage that is critical to the pilotage functions. Note that the LWIR horizon contrast is typically much higher than in MWIR, but the main contributor to this large contrast is the path radiance of the sky (which is always much higher in equivalent blackbody temperature in the MWIR). In fact, sometimes, the LWIR sky-to-ground contrast is a problem in pilotage in that it used a large portion of the scene dynamic range, making it hard to see the smaller contrast features in a scene. In some pilotage systems, dynamic range compression techniques are used to reduce the sky-to-ground contrast in the presented imagery.

There is some work verifying the importance of SCT. At RDECOM CERDEC Night Vision and Electronic Sensors Directorate,¹⁰ researchers conducted human perception experiments to verify different metrics associated with SCT. The goal of the experiment was to compare images with different SCT and measure the perceived image quality. Experimenters adjusted the gain and level of different image sets, and mapped target contrast and TTP (these metrics directly correlate with SCT) against perceived image quality. The data show that perceived image quality rises as target contrast and TTP are increased, and quickly falls off as features in the scene begin to saturate. Note that as SCT rises, target contrast will fall and TTP will rise.

The contributions associated with σ_T and SCT can be quantified in three different ways. First, it is worth describing the actual detected signal is due to a combination of actual temperature of the objects in the scene as well as each of the object emissivities. Each object in the scene has an actual temperature and an actual emissivity. Higher temperature objects emit more power and photons, and higher emissivity objects emit more power and photons. However, a radiometric term called equivalent blackbody temperature is used to describe the flux coming from an object as the same flux provided by a blackbody (with emissivity of 1) at an equivalent blackbody temperature. The equivalent blackbody temperature of an object accounts for both the actual temperature and the emissivity of the object.

The three ways SCT can be described above are equivalent blackbody temperature (as it is reported in this paper), radiometry in watts (W), and radiometry in photons/second (p/s). Typical equivalent blackbody temperatures for the average background and a typical target contrast of 1K are given in Table 2.

For terrestrial scenes, the typical background in equivalent blackbody temperature is 300K for both MWIR and LWIR. A typical target or object differential contrast can be around 1K in both MWIR and LWIR. These quantities are useful when using a camera with a noise equivalent temperature difference (NETD), as given in the table (e.g., 20 mK), so that a quick signal-to-noise (SNR) ratio is just the target contrast divided by the NETD, which for Table 2 is $\text{SNR} = 50$.

Table 2 SCT radiometric parameters.

Parameter	LWIR	MWIR
Typical background temperature, T_{bg}	300K	300K
Typical target contrast, dT_{tgt}	1K	1K
Typical sensor NETD	20 mK	20 mK
Background emittance W/cm^2 , M_{bgw}	1.202E-02 W/cm^2	6.21E-4 W/cm^2
Target contrast W/cm^2 , dM_{tgtw}	1.96E-4 W/cm^2	2.27E-5 W/cm^2
Background emittance $p/s\text{-}cm^2$, M_{bgp}	4.90E17 $p/s\text{-}cm^2$	1.13E16 $p/s\text{-}cm^2$
Target contrast $p/s\text{-}cm^2$, dM_{tgtp}	7.88E15 $p/s\text{-}cm^2$	4.08E14 $p/s\text{-}cm^2$

The SCT equivalent blackbody temperatures for the background and object contrast (e.g., SCT) can be converted to a “watt” based radiometry. These quantities are useful when working with power detectors or detectors that are specified with detectivity. The typical background and target contrast values in Table 2 are converted to source emittance given in W/cm^2 . There are two important things to note here. First, the LWIR target contrast rides on a large background temperature power pedestal where the target contrast is only a few percent of the overall power received from the average background signal. Second, the LWIR has 8.6 times the target contrast in radiometric power than the MWIR for the same SCT.

For those working with photon counting detectors and cameras, the equivalent blackbody temperatures for the backgrounds and targets/objects can be converted to photon radiometry in emittance. Table 1 notes the pedestal of the background as well as the target contrasts in the MWIR versus LWIR in $p/s\text{-}cm^2$. MWIR detectors usually have a smaller capacitor well for collection photo electrons due to the smaller background and contrast. In this case, the number of photons collected in the LWIR is 19.3 times the photons collected in the MWIR. The reason for the larger ratio is that, on average, each photon per second in the MWIR requires more energy (and corresponding power) to generate than in the LWIR by a factor of 2.5.

Planck’s curve for a 300K equivalent blackbody background is shown in Fig. 3. The emittances shown explain the reasons for power in watts and power in photons/second are much larger in the LWIR than in the MWIR. For a given constant SCT near terrestrial temperatures, the contrast power in watts is just under 10× MWIR versus LWIR. And, in photons per second, the ratio is just under 20× MWIR versus LWIR. This is the reason that MWIR sensors for the same camera design (FOV, detector size, optics, etc.) require a detector integration time of between 10× and 20× longer than LWIR to achieve the same SNR.

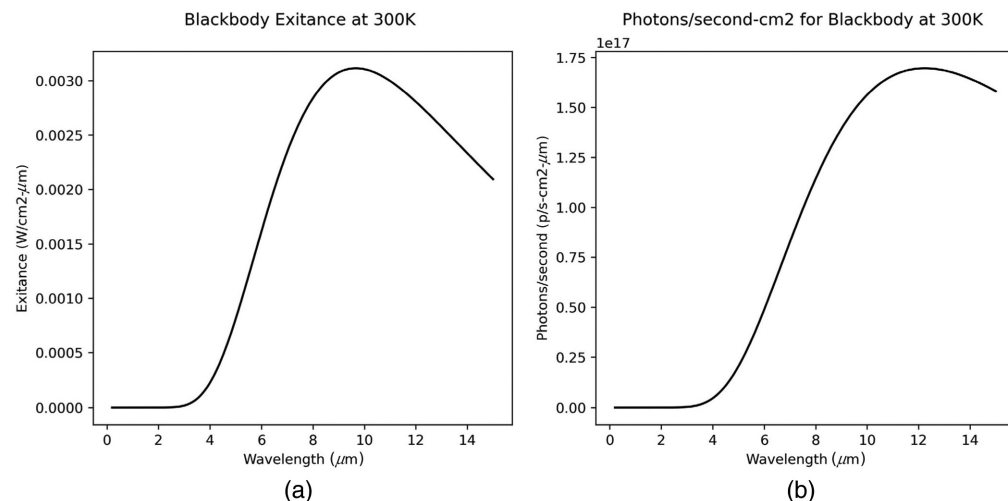
**Fig. 3** Planck’s curve for emittance from a 300K equivalent blackbody in W/cm^2 and $p/s\text{-}cm^2$.

Table 3 System information.

	FLIR T1020	TEL-7958
Band	8 to 12 μm (LWIR)	3 to 5 μm (MWIR)
Format	1024 $\times$ 768	640 $\times$ 512
Pitch	17 μm	15 μm
FOV	34 deg $\times$ 24 deg	26 deg $\times$ 21 deg
IFOV	0.58 mrad	0.71 mrad
NETD	<20 mK @ 30°C	<20 mK @ 25°C
F/#	$f/1.2$	$f/2.3$

While measurements of σ_T are reported in Kelvin associated with equivalent blackbody temperature, keep in mind the differences associated with the signals in watts and in photons/second. The measurements provided here can be easily converted using Planck's equation.

3 Experimental Description

3.1 Sensors

Two cameras were used to obtain the MWIR and LWIR data: a Telops-7958 MWIR camera and a FLIR T1020 LWIR camera. The camera parameters are provided in Table 3. The cameras were converted to radiometers using a large area blackbody calibration source. To compare scene contrast between MWIR and LWIR, imagery was gathered in multiple scenes from the same perspective in both bands. The output of both cameras was set to be linear with temperature.

Since the FOVs were different, only the same area in the image was selected in the analysis. At longer ranges, the difference in IFOV can influence the SCT measurement, but all the measurements provided in this research are at close range.

3.2 Calibration

Images were taken by the FLIR T1020 and TEL-7958 mounted adjacently on the same tripod to minimize parallax. The pointing direction of the imagers was the same since the two FOVs were aligned. Auto-exposure and auto gain were disabled, and the dynamic range was set between the absolute maximum and minimum equivalent blackbody temperatures contained in the scene.

Before taking scene imagery, the two sensors were calibrated such that they both had the same baseline for equivalent blackbody temperature. Two programmable blackbodies (model BB6 from True Colors Infrared Imaging, 15 $\times$ 15 cm square apertures) were positioned such that they were both in the FOV of the MWIR and LWIR cameras and were set to temperatures near the highest and lowest equivalent blackbody temperatures contained in the scene. The blackbodies used have a typical emissivity of above 0.98; therefore, an emissivity of 1 is used and the maximum error associated with this imperfect emissivity is calculated. The maximum (for an emissivity of 0.98) error in the equivalent blackbody temperature of the blackbody source is 0.44% of the absolute temperature in LWIR, and 0.19% of the absolute temperature in MWIR. The error in the temperature of the blackbodies propagates to a σ_T error in each data set. For a scene with $\sigma_T = 3$, the maximum error in σ_T is 0.8% in LWIR and 0.34% in MWIR.

Each possible gray level in both LWIR and MWIR images were plotted along their respective equivalent blackbody temperatures. The relationship provided in Fig. 4 was considered the system intensity transfer function (SITF).

Data from the measured scenes were exported into ython in.jpeg (8-bit) format, and areas-of-interest were chosen such that the content of the LWIR scenes match the content of the MWIR scenes. Matching the content between the LWIR and MWIR scene by choosing a similar areas-of-interest is likely the largest source of error in this study, as there is parallax between the two images, as well as differences in FOV. For a set of sample images, attempting to choose the same

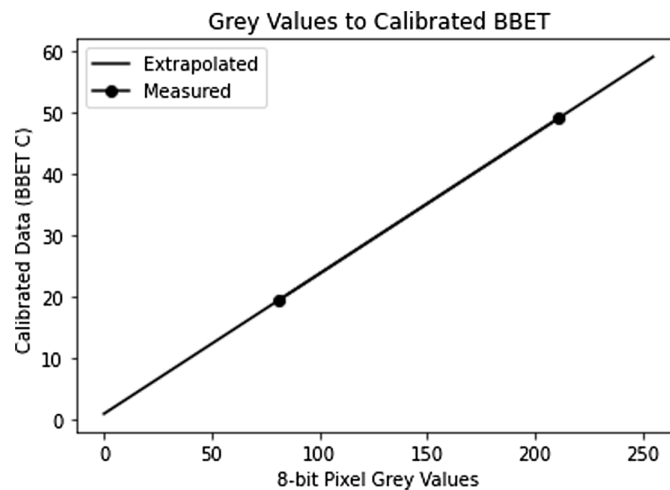


Fig. 4 Linear relationship of equivalent blackbody temperature to gray level.

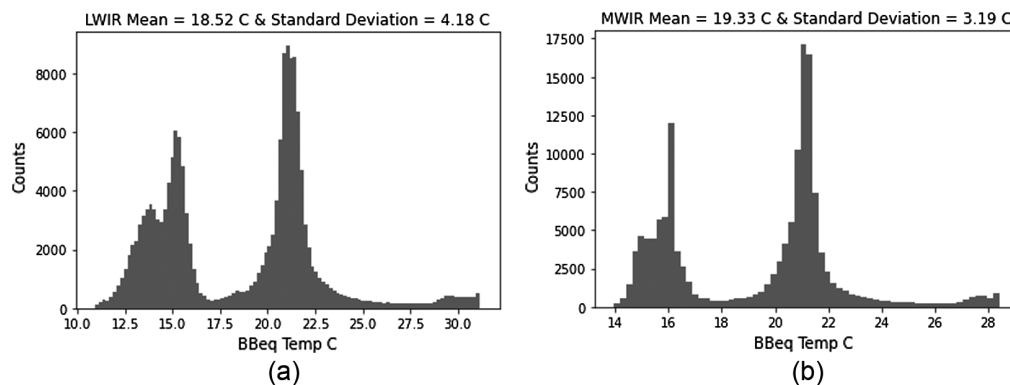


Fig. 5 Resulting histograms from analysis of a single scene comparison between LWIR (a) and MWIR (b).

areas-of-interest over multiple trials created an average of a 1% error in σ_T . Great care is taken with each image to minimize this error such that it does not affect the overall trend of the data.

Each area-of-interest is converted into a list of pixel values, and the SITF is applied to the pixel values such that they are converted into equivalent blackbody temperatures, resulting in a list of equivalent blackbody temperatures. These lists are used to calculate the σ_T of each scene. Figure 5 shows the two resulting histograms of the conversion of a scene into a list of equivalent blackbody temperatures. These histograms illustrate that there are slight differences between the different bands in each scene statistically, and these slight differences are what drive the difference in σ_T and ultimately SCT.

3.3 Conditions

Data were collected in both rural and urban scenes to get a wide variety of content. Rural scene included vegetation and rock formations, whereas urban data included a wide variety of man-made structures and objects. For each rural and urban scene, nighttime and daytime imagery were collected for both the LWIR and MWIR. Data were also collected on a cloudy day so that it could be compared to data collected on a clear day.

To further investigate the effects of daytime, a diurnal experiment was conducted where the FLIR T1020 and TEL-7958 acquired images every 5 min for 24 h. This experiment occurred on a clear day, with the blackbodies in the FOV of the imager for each image taken. The imagery was taken around Tucson, Arizona where the climate was high desert in the month of May. For all data, the ambient temperature was 30°C and humidity was low, below 25%.

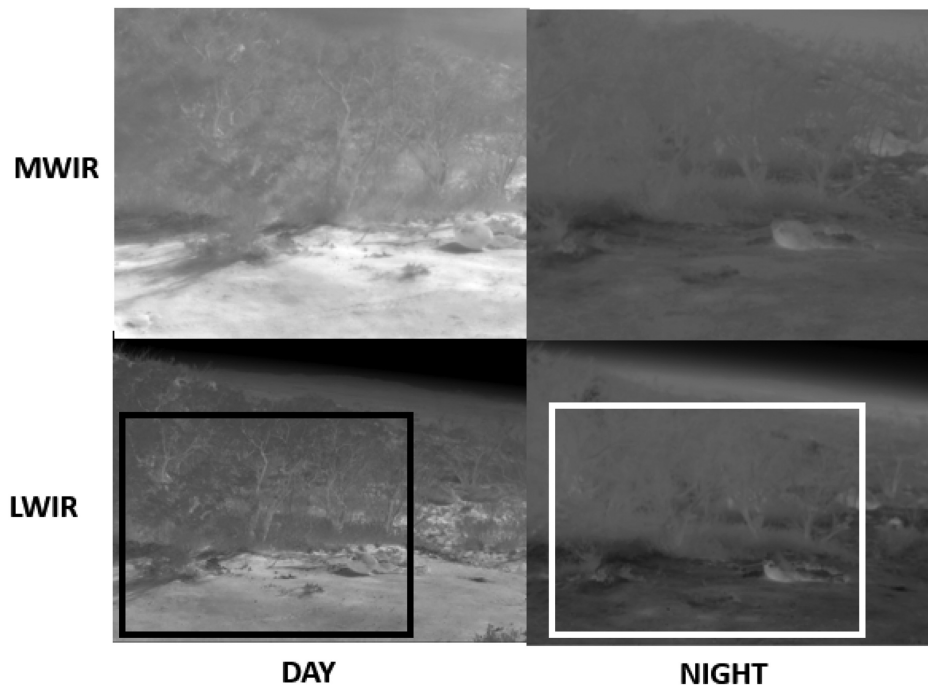


Fig. 6 Sample rural imagery [MWIR (top) and LWIR (bottom); daytime (left) and nighttime (right)]. Boxes indicate area analyzed.

4 Data and Analysis

4.1 Rural Data

Sample images of scenes from the rural data collection are shown in Fig. 6. The top images are MWIR, and the bottom images are LWIR. On the left side are images taken at daytime and on the right side are images taken at nighttime. It is worth noting that the dynamic range of the sensors is set in the scene such that no pixels are saturated. Boxes are drawn to illustrate the FOV of the LWIR used in analysis to provide the same scene content as the MWIR sensor.

Thirty sets of four images were analyzed in rural conditions. The four images included both MWIR and LWIR imagery as well as day and night conditions. The σ_T of the scene in the MWIR against the σ_T in the LWIR for each MWIR/LWIR pair is plotted and shown in Fig. 7. The

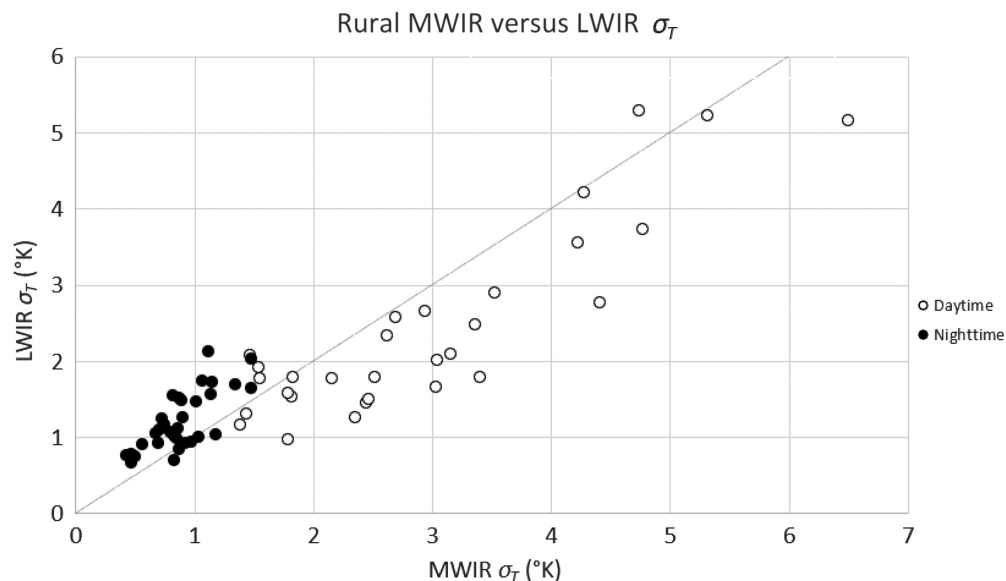


Fig. 7 Rural scene contrast distribution for daytime and nighttime.

diagonal line provides a plot location where the MWIR and LWIR signatures are equivalent. The data points above the line corresponded to an advantage in LWIR contrast and data points below this line corresponded to an advantage in MWIR contrast.

4.2 Urban Data

Sample images of the urban scenes are shown in Fig. 8. The top images are MWIR, and the bottom images are LWIR. The left side are images taken at daytime and the right side are images taken at nighttime. Boxes are drawn to illustrate the FOV of the images used in analysis and correspond to the same regions of the image in both the MWIR and LWIR. The scenes collected have many urban objects to include manmade park features, buildings, roads, windows, lamp poles, etc. Some objects in the urban scenes included thermal variations from the rural scenes since buildings have air conditioners and have other manmade structures to control thermal conditions. All urban scenes are outside; no inside imagery is used.

The day and night results are plotted in Fig. 9 for both MWIR and LWIR. The day results are plotted as the blue dots and the night results are plotted as the orange triangles. The diagonal

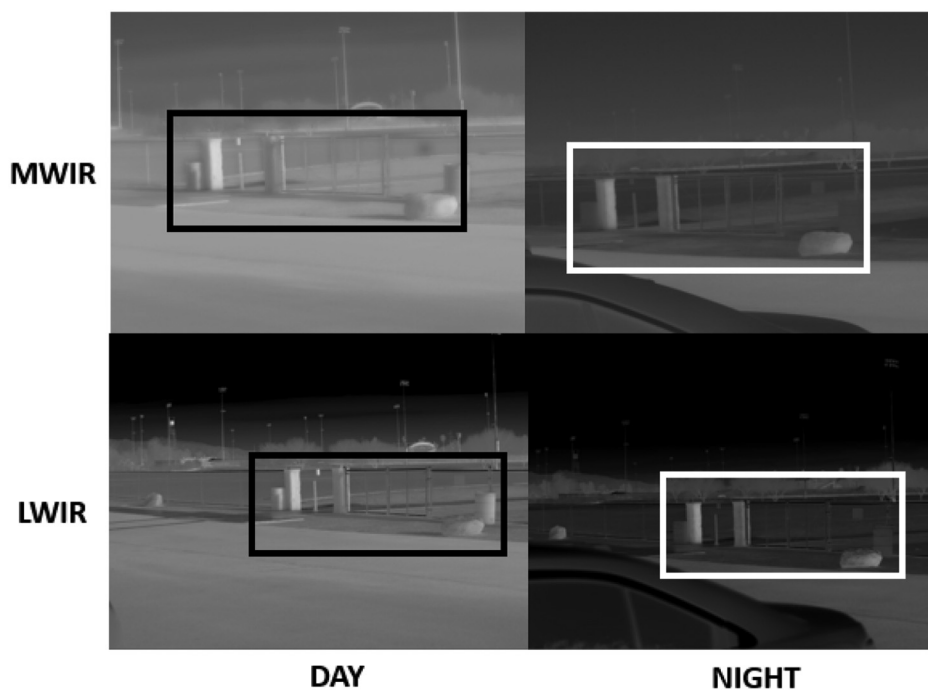


Fig. 8 Sample urban imagery [MWIR (top) and LWIR (bottom); daytime (left) and nighttime (right)]. Boxes indicate area analyzed.

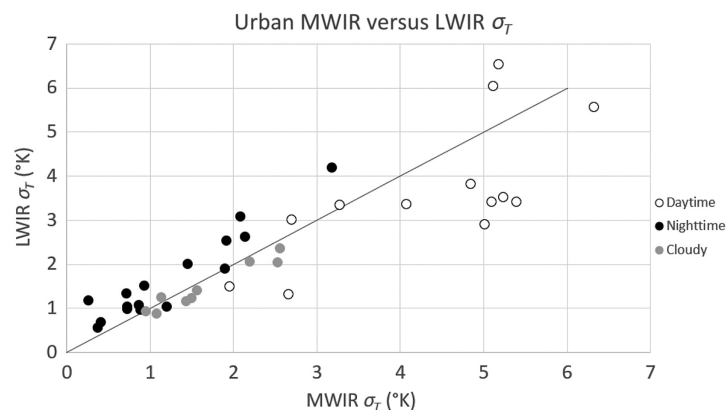


Fig. 9 Urban scene contrast distribution for daytime and nighttime.

reference is also included in the plot. There was also an opportunity to obtain daytime results with heavy cloud overcast one day and we obtained the urban data for cloud overcast conditions. These data are plotted as the gray squares in Fig. 8.

4.3 Diurnal

The SCT and σ_T change with ambient temperature, Sun angle, time, and the attempt of all objects in the scene to achieve thermal equilibrium. The Sun irradiance and the Sun angle have a large effect on both LWIR and MWIR scene contrast due to objects differentially heating up with different amounts of solar illumination and differences in absorptivity. The result of large variations in SCT due to absorbed sunlight is sometimes called solar loading. In addition to solar loading, a significant amount of contrast in the MWIR is due to solar reflections with differential reflectivities and differential amounts of solar illumination reaching different objects. In the LWIR, almost all flux seen during the day is due to the emitted light. In the MWIR, the flux seen during the day includes reflected solar illumination, which can be a significant contribution to the overall SCT. Reference 3 shows for a sandy surface, the MWIR solar reflections can be 20% to 30% of the SCT signature.

To provide a measure of the diurnal effects, the FLIR T1020 and TEL-7958 were set up to take an image of the same scene every 5 min for 24 h. Samples of the diurnal scenes are shown in Fig. 10. The top images were MWIR, and the bottom images were LWIR. The left side images were taken at daytime and on the right side were images taken at nighttime. Boxes show the areas that were used in the image analyses. The camera positions were static, and the areas taken in the MWIR and LWIR corresponded to the same scene area.

Figure 11 shows the σ_T for MWIR and LWIR over a period of 24 h for the scenes shown in Fig. 10. The diurnal cycle started and ended at 3:20 p.m. The orange plot corresponds to the MWIR σ_T and the dashed blue dots correspond to the LWIR σ_T . The data were taken from April 8 to April 9 in Tucson, Arizona, and the conditions were sunny, clear sky, and dry. The maximum temperature on April 8 was 29°C and the maximum temperature on April 9 was 31°C. The humidity ranged between 9% and 32%.

Since the night data are difficult to see in Fig. 11, we provide more detail for the night plot in Fig. 12. LWIR σ_T is always higher than MWIR σ_T , where the ratio of MWIR σ_T to LWIR σ_T is roughly 2/3rds through the night. The other aspect of night σ_T is that for both MWIR and LWIR it seems to decrease as the heat of the objects bleeds off toward ambient as the night continues. The lowest σ_T appears just before dawn.

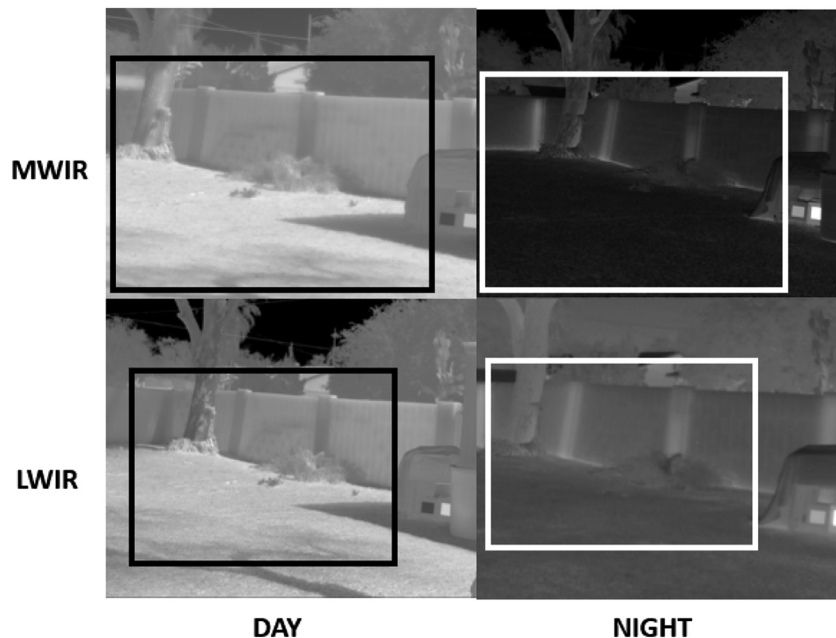


Fig. 10 Sample diurnal imagery [MWIR (top) and LWIR (bottom); daytime (left) and nighttime (right)]. Boxes show area analyzed.

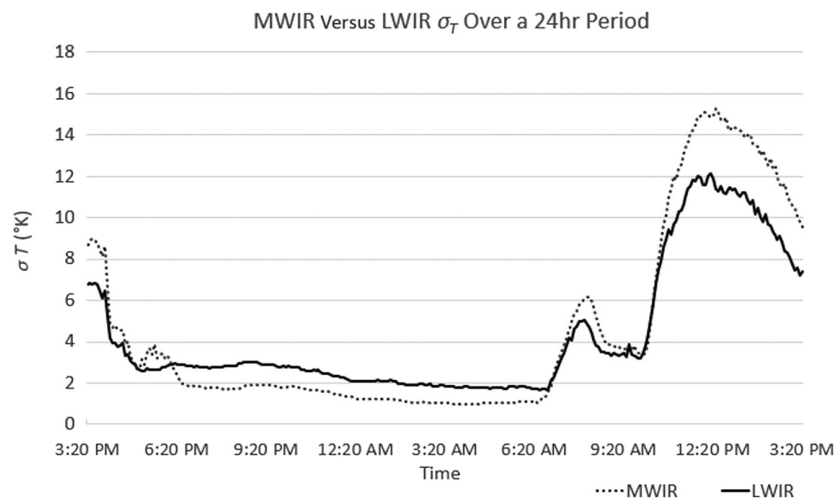


Fig. 11 MWIR versus LWIR scene contrast over a 24-h diurnal cycle.

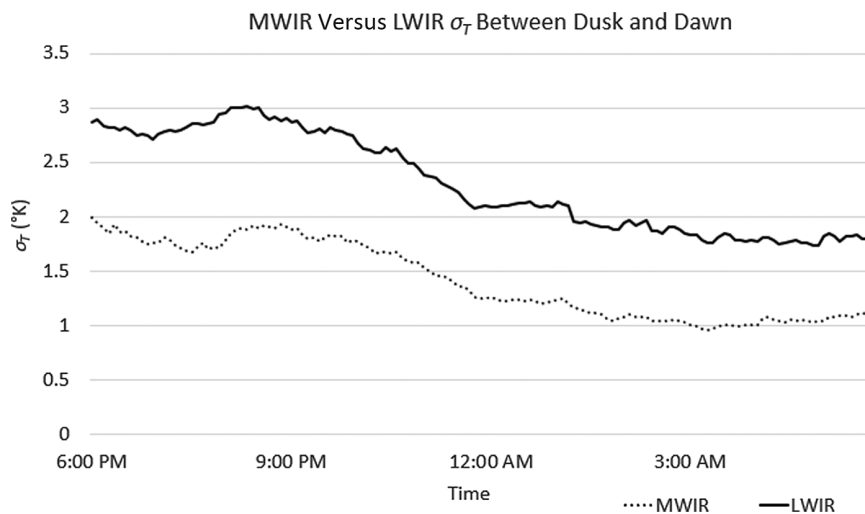


Fig. 12 MWIR versus LWIR nighttime scene contrast. This is a “zoomed” view of Fig. 11.

5 Discussion

Time of day, climate, environment, temperature, conditions, and spectral band all play a large role in both MWIR and LWIR SCT and σ_T . Time of day and year also influences the amount of solar radiation striking the objects in the scene. This radiation differentially heats up objects with different emissivities, reflectivities, and thermal masses resulting in different SCTs and σ_T . Reflectivity and emissivity are a function of wavelength and therefore also vary between MWIR and LWIR.

Data from rural scenes indicated a clear difference between MWIR and LWIR σ_T at different times of day. In almost all daytime images analyzed, MWIR had a higher scene contrast than LWIR due to the additional solar reflections in the MWIR that contributed to the SCT and σ_T . The ratio in MWIR/LWIR during the day, depending on how it was calculated, is roughly 1.1 to 1.2. The day σ_T ranged from 2 to 6.5K in the MWIR and 1 to 6.5K in the LWIR. At night, the LWIR σ_T was almost always larger than the MWIR σ_T . The night σ_T was significantly lower than day σ_T . The night σ_T ranged from 0.4 to 1.5K in the MWIR and 0.6 to 2K in the LWIR.

In the urban environment, there were many man-made materials and structures that had even more emissivity and reflectivity variations than rural scene objects. The data were spread more in the urban environment than the rural data, as expected. However, the overall trends were the

same. Day σ_T favored the MWIR slightly and night σ_T favored the LWIR. The night MWIR σ_T was generally slightly lower than LWIR σ_T by roughly the same ratio seen in the urban environment. The day σ_T ranges were about the same as the urban σ_T and the night σ_T were higher than day σ_T . One reason for this was the hot objects associated with buildings and infrastructure as well as windows that reflected the cold sky. One interesting note was that when the day measurements were made with cloud cover, the LWIR and MWIR σ_T were the same, suggesting that the reflectivities that allowed the cold sky to contribute were then reflecting clouds that were at the same equivalent blackbody temperatures in the MWIR and LWIR. The cloud-covered day σ_T were not as cold as the night but not as high as sunny day σ_T .

The diurnal data supported both the rural and urban data results. The day σ_T were similar and the night σ_T were also similar. Day σ_T were higher in the MWIR, and night σ_T were higher in the LWIR. The ratio of MWIR/LWIR night σ_T was about 2/3rds for the entire night. The night σ_T ranged from 1 to 2K in the MWIR and from 1.7 to 3K in the LWIR. In both MWIR and LWIR cases, the σ_T reduced as the night progressed and was lowest just before dawn.

The data and analysis results mean that the scene clutter is much lower during the night, so search and detection with hot vehicles and humans at night is much easier than the same task during the day in both the MWIR and LWIR bands. Also, at night, clutter is smaller in the MWIR so searching for vehicles or humans at night would be easier in the MWIR, especially in the rural environment. For pilotage and scene-based navigation at night, the LWIR has more contrast, so LWIR would be preferred. Depending on the type of detector used, the LWIR provides roughly 15 times the power scene contrast of the MWIR, and the LWIR provides roughly 30 times the photon counting contrast of the MWIR.

6 Conclusion

The data and analysis presented suggest that MWIR has higher σ_T and SCT than LWIR in the daytime with sunny conditions. They are about the same in cloud covered conditions. At night, MWIR has less σ_T and SCT than LWIR by a rough factor of 2/3rds. For pilotage and scene-based navigation, in environments with low scene contrast, slight improvements in scene contrast matter more than ever, meaning LWIR may be a safer choice. While the night differences in SCT are not significant (roughly 2/3rds), there may be other reasons to consider MWIR systems. For targeting, it appears that a benefit for MWIR is that the clutter is lower in SCT. The clutter SCT (σ_T) in the LWIR is higher and may compete with target detection more.

In future work, LWIR and MWIR scene contrast should be investigated at the horizon due to the differences in sky path radiance. Other work should include SCT and σ_T under a different climate than Tucson such as hot and humid (like Florida). Finally, SCT and σ_T should be analyzed in extremely low contrast conditions, such as rain and post-cold-rain conditions.

Disclosures

There are no conflicts of interest for any authors listed in this paper.

Code, Data, and Materials Availability

All data in support of the findings in this paper are available within the article.

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Biographies of the authors are not available.